CONVERTING FORTRAN IN TO A MORE MODERN PROGRAMMING LANGUAGE: A JAVA CONVERSION

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Converting Fortran into another programming language required analysis of how the source program worked. The program simply utilized a class like object in Fortran to establish a weather type. All computations were performed in the weather type and they were simply called upon in the main.f90 code. The language that was chosen was Java. This was for several reasons. First off, due to the nature of the program itself, it was evident that compiled code was more advantageous instead of interpreted languages from a speed of execution standpoint. Additionally, since Java is implemented using a Java Runtime Environment it instantly gains portability for use in a multitude of environments without needing to pay particular attention to host operating systems or underlying hardware itself. Additionally, given the robust support for Java and its ability to run on mobile code, this makes the software very adaptable for a multitude of use cases.

Several alternative languages were considered but coupled with the reasons above and further exploration it was evident that Java is the clear preferred choice. The first language that was considered was Python. While Python is very scalable with thousands of libraries that are developed in all sectors, it does suffer in terms of speed of execution at run time. Considering the critical nature of the calculations that a weather prediction system for rocket launches requires an interpreted language such as Python would not be the best option. The second language that was considered was C++. Generally, C++ is faster than Java, the difference is marginal. The real reason that C++ was not considered was due to the complexity of the language in comparison to Java and the fact that often, the source code would have to be recompiled for every platform it is deployed to. This can create a significant bottleneck on the stance of pushing updates to all devices running the software. Simply put, Java provides many of the benefits of the other languages, without too many of the drawbacks and that is why it was chosen for this application.

Conversion was performed using a guided AI (ChatGPT) method. The code was analyzed line by line and each step of the process was verified to be understood by the AI. The next step was to have the AI replicate the code for use. The AI was tasked with explaining the code for the new Java version code. Verification of the code was performed by this analysis and by compiling the original Fortran code and the Java code and comparing output as well as verifying the underlying logic works in performing the actions originally required of the first code.

Testing of the code was automated using Junit 5. The testing plan used was as follows:

Test Case	Input	Expected Output	Pass/Fail
Initialization	Arrays from example	Weather object created with correct fields	Pass
Today	Data from program	Sunday's forecast	Pass
Averages	fh = {10, 10, 10, 10, 10, 10, 10}	10.0	Pass
Averages high	fh = {78, 76, 80, 82, 85, 79, 75}	79.2857	Pass
Averages Low	fl = {75, 70, 75, 76, 75, 70, 69}	72.8571	Pass
Weekly High	fh = {78, 76, 80, 82, 85, 79, 75}	85	Pass
Weekly Low	fl = {75, 70, 75, 76, 75, 70, 69}	69	Pass
Description	wc = 'S'	"SUNNY"	Pass
Description	wc = 'X'	"UNKNOWN"	Pass
Weekly Output	Data from program	Per-day output matches expected	Pass

Testing was performed with test cases using Junit and verifying default program output. Junit was simply used because it is the de-facto industry standard and is automated, repeatable and extremely fast. It also easily integrates into Visual Code Studio, the chosen IDE for the project. Overall, the program converted very nicely and efficiently into the chosen language. While larger code samples may be more difficult for AI to convert, this project is a clear example of how useful this tool can be in the realm of software development and portability.